Beamline for Schools 2026

Experiment Proposal

Dual-Readout Calorimetry:

Separating Electromagnetic and Hadronic Shower Components

Using Simultaneous Scintillation and Cherenkov Light Measurement

The Dualists

TED Ege College, Turkey

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1. Motivation

1.1 Why We Want to Participate

We are a team of 10 high school students from TED Ege College, Turkey, who share a deep curiosity about how physicists detect and measure particles. When we learned about the Beamline for Schools competition, we saw an extraordinary opportunity to experience real experimental physics firsthand. Our team includes students interested in physics, engineering, and data analysis, and we believe that this experiment will challenge us to combine all these skills.

Turkey has a growing presence in international particle physics through its contributions to the CMS experiment at the LHC. We are inspired by this legacy and want to contribute to the next chapter of detector development. Participating in BL4S would not only be an incredible scientific experience but also motivate other students in Turkey to pursue careers in fundamental research.

1.2 Scientific Motivation

One of the biggest challenges in modern particle physics is accurately measuring the energy of hadrons (particles like pions and protons). While electromagnetic calorimeters can measure electron and photon energies with excellent precision, hadronic calorimeters suffer from a fundamental problem: the **invisible energy** lost in nuclear interactions.

When a hadron enters a calorimeter, it creates a hadronic shower containing two components: an electromagnetic (EM) part coming from neutral pion decay into photon pairs, and a hadronic part involving nuclear breakup, slow neutrons, and nuclear excitation. The hadronic part deposits less visible energy than the EM part, and critically, the EM fraction fluctuates randomly from event to event. This fluctuation is the dominant source of poor energy resolution in hadron calorimeters.

The **dual-readout** technique offers an elegant solution to this problem. By simultaneously measuring two different types of light from the same shower, one can determine the EM fraction on an event-by-event basis and correct the energy measurement accordingly. This principle is at the heart of the IDEA detector concept proposed for the Future Circular Collider (FCC-ee) at CERN, making it one of the most important R&D topics in detector physics today.

Our experiment aims to demonstrate this dual-readout principle using equipment available at the BL4S facility at CERN.

2. Physics Background

2.1 EM vs Hadronic Showers

When a high-energy electron enters dense material, it produces an electromagnetic shower through bremsstrahlung and pair production. All particles in this shower are relativistic (electrons, positrons, photons), so the shower produces both scintillation light (from ionisation energy loss) and Cherenkov light (from charged particles exceeding the speed of light in the medium). The ratio of Cherenkov to scintillation light (C/S) is essentially constant for EM showers.

A hadronic shower is fundamentally different. When a pion or proton undergoes a nuclear interaction, the shower contains: (a) an EM component from neutral pion decays, which behaves like a small EM shower, and (b) a hadronic component including slow protons, neutrons, nuclear fragments, and nuclear de-excitation. The slow, heavy particles in the hadronic component produce scintillation light but are below the Cherenkov threshold. Therefore, the C/S ratio is lower for hadronic showers and varies event-by-event depending on the random EM fraction.

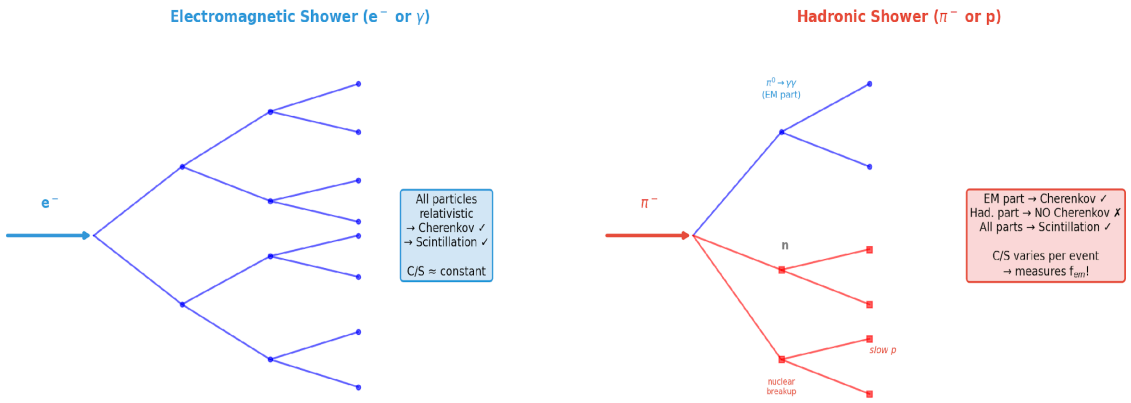

Figure 1: Comparison of electromagnetic (left) and hadronic (right) showers. In EM showers, all particles emit both Cherenkov and scintillation light (C/S constant). In hadronic showers, the slow hadronic component produces scintillation but no Cherenkov light (C/S varies).

2.2 The Dual-Readout Principle

The key insight is that by measuring both the scintillation signal S and the Cherenkov signal C for each event, we can extract the EM fraction. For a shower with EM fraction f_em, the two signals can be written as:

$$S = E \times [f_em + (h/e)_S \times (1 - f_em)]$$

$$C = E \times [f_em + (h/e)_C \times (1 - f_em)]$$

where (h/e)_S and (h/e)_C are the hadronic-to-electromagnetic response ratios for scintillation and Cherenkov light respectively. Since (h/e)_C is much smaller than (h/e)_S (because slow hadrons produce scintillation but almost no Cherenkov light), the two equations can be solved simultaneously to obtain both E and f_em for each event. This event-by-event correction dramatically improves the hadronic energy resolution.

2.3 Connection to FCC-ee and IDEA

The dual-readout approach is the core technology of the IDEA (Innovative Detector for an Electron-positron Accelerator) detector, proposed for the FCC-ee, CERN's next-generation electron-positron collider. In IDEA, alternating scintillating and clear (Cherenkov) fibres are embedded in a metal absorber. The RD52/DREAM collaboration at CERN has been developing this concept for over a decade. Our experiment tests the same fundamental principle in a simplified configuration accessible at BL4S, providing a direct link to cutting-edge detector R&D.

3. Experimental Setup

3.1 Overview

Our experiment uses the CERN PS T9 mixed secondary beam, which contains electrons, pions, and protons at momenta between 0.5 and 10 GeV/c. The particle type is identified event-by-event using the threshold Cherenkov counter (XCET). Each particle then passes through our dual-readout calorimeter module, which consists of interleaved layers of lead-glass blocks (providing the Cherenkov signal C) and plastic scintillator tiles (providing the scintillation signal S).

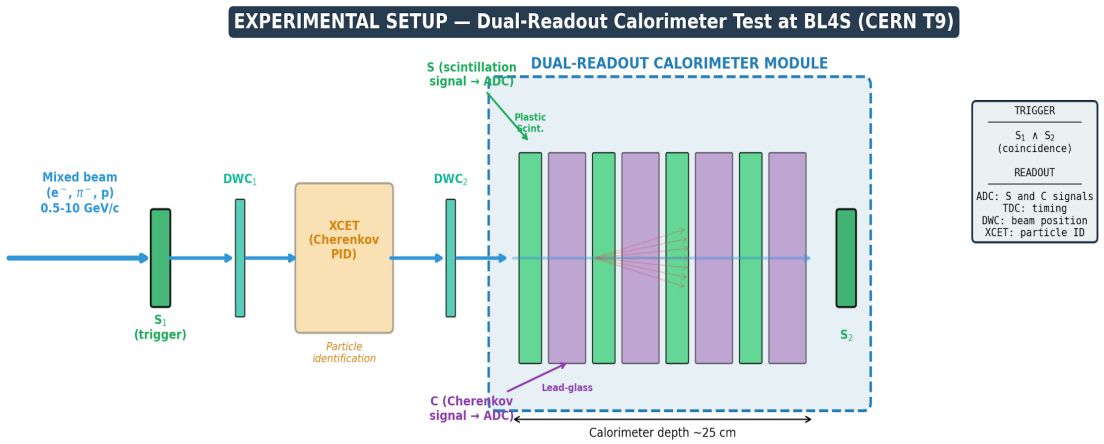

Figure 2: Experimental setup showing the beam entering from the left, passing through trigger scintillators (S1, S2), tracking chambers (DWC1, DWC2), and the threshold Cherenkov counter (XCET) for particle identification, before entering the dual-readout calorimeter module consisting of alternating plastic scintillator and lead-glass layers.

3.2 Detector Components

Trigger scintillators (S1, S2): Two plastic scintillators placed before and after the calorimeter provide the trigger signal via coincidence (S1 $\wedge$ S2). This ensures that only particles passing through the full setup are recorded.

Delay Wire Chambers (DWC1, DWC2): Two DWCs measure the beam position with ~200 µm resolution. This allows us to select events where the beam hits the calorimeter at a well-defined position, reducing edge effects.

Threshold Cherenkov Counter (XCET): The gas-filled Cherenkov counter at CERN allows particle identification by adjusting the gas pressure. At a given momentum, different particle species have different velocities and thus different Cherenkov thresholds. By setting two pressure points, we can separate electrons, pions, and protons event-by-event. This is our ground truth for validating the dual-readout separation.

Dual-readout calorimeter module: This is the core of our experiment. We propose to use the lead-glass blocks already available at CERN T9 (which produce Cherenkov light read out by photomultipliers) interleaved with plastic scintillator tiles (which produce scintillation light, also read out by photomultipliers). Both signals are digitised by ADCs. For each event, we record the pair (S, C). The total depth should be approximately 20–25 cm, corresponding to roughly 1 nuclear interaction length for pions.

3.3 Trigger and Data Acquisition

The trigger is formed by the coincidence of S1 and S2. Each triggered event records: the ADC values for all scintillator and lead-glass channels (giving S and C), the TDC values for timing, the DWC hit positions for beam tracking, and the XCET status for particle identification. At a beam rate of up to 3000 particles per second, we can collect millions of events over the two-week experimental period.

4. Measurement Plan

4.1 Data Collection Strategy

We plan to take data at several beam momenta: 1, 2, 3, 5, 7, and 10 GeV/c. At each momentum, we will use both positive and negative beam polarities to access different particle compositions. The XCET pressure will be adjusted at each momentum to optimize particle identification. For each configuration, we aim to collect at least 50,000 events, which takes approximately 20 seconds at full beam rate.

4.2 Analysis Steps

Step 1 — Particle identification: Using the XCET signal and TOF information from S1 and S2, we tag each event as electron, pion, or proton.

Step 2 — C vs S scatter plot: For each beam momentum, we plot the Cherenkov signal C versus the scintillation signal S for all events. We expect to see distinct populations for electrons (high C/S, tight cluster) and hadrons (lower C/S, broader distribution), as shown in our simulated results below.

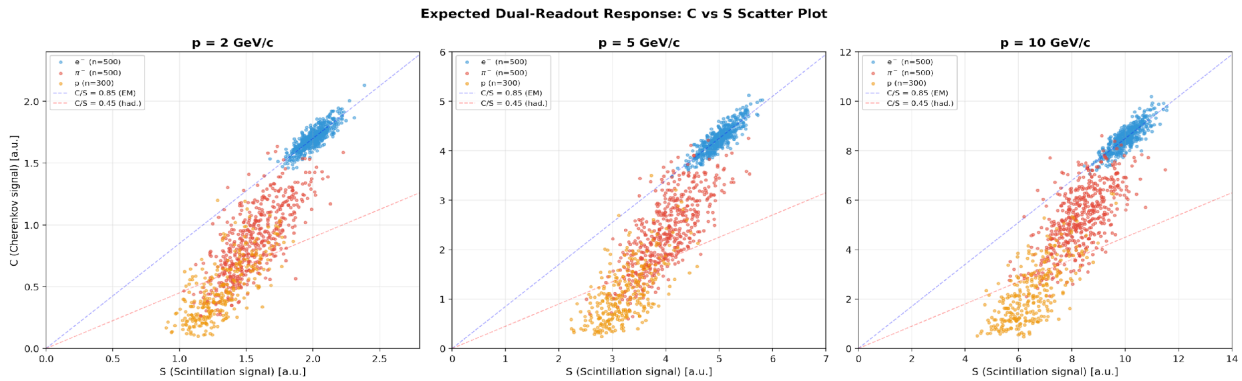

Figure 3: Simulated C vs S scatter plots at three beam momenta. Electrons (blue) form a tight cluster with constant C/S ratio. Pions (red) and protons (orange) show lower C/S with event-by-event fluctuations reflecting the varying electromagnetic fraction in hadronic showers.

Step 3 — EM fraction extraction: From the C/S ratio of each hadron event, we extract the EM fraction f_em using the calibration obtained from the pure electron sample (which defines the C/S ratio for f_em = 1).

Step 4 — Energy correction: Using f_em, we apply the dual-readout correction formula to obtain the corrected energy for each hadron event. We then compare the energy resolution before and after correction.

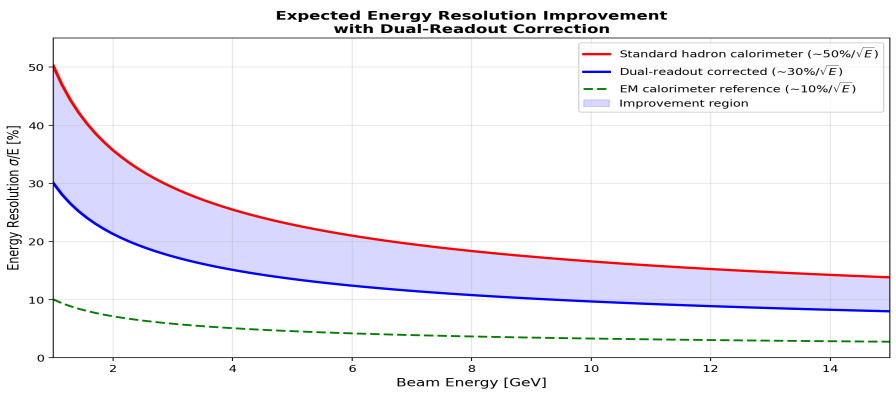

Figure 4: Expected improvement in hadronic energy resolution after dual-readout correction. The standard resolution (red) is dominated by EM fraction fluctuations. The corrected resolution (blue) approaches the intrinsic stochastic limit.

Step 5 — Energy dependence: We repeat the analysis at each beam momentum to study how the dual-readout improvement depends on energy. At higher energies, the EM fraction increases (logarithmically), which should be directly visible in our C/S distributions.

5. Expected Results and Feasibility

5.1 Expected Observations

We expect to observe the following:

1. Clear separation between electron and hadron populations in the C vs S plane at all beam momenta.
2. The electron C/S ratio should be constant across energies (pure EM showers).
3. The pion C/S distribution should shift upward with increasing energy (higher average EM fraction at higher energies due to multiplicity effects).
4. The dual-readout correction should visibly narrow the hadron energy distribution.
5. Protons should show a distinct C/S signature compared to pions due to different shower development.

5.2 Feasibility Assessment

Parameter	Value / Status
Beam type	CERN PS T9 mixed beam (e, π, K, p)
Beam momentum	0.5 – 10 GeV/c (user selectable)
Beam rate	Up to 3000 particles/s
Lead-glass blocks	Available at CERN T9 (3×3 matrix)
Plastic scintillators	Available at CERN T9
XCET (Cherenkov PID)	Available, pressure adjustable
DWC tracking	Available (200 μm resolution)
Readout	CAEN ADC (V792) and TDC (V1290)
Events needed	~50k per momentum (< 1 min)
Total beam time	< 1 day for full programme
Own equipment needed	None — all equipment provided by BL4S

A critical strength of our proposal is that **all required equipment is already available at the CERN T9 area.** We do not need to bring any custom detectors. The lead-glass calorimeter, plastic scintillators, DWCs, Cherenkov counters, and readout electronics are all part of the standard BL4S equipment. This makes our experiment highly feasible and low-risk.

5.3 Potential Challenges

Shower containment: With ~25 cm of material, some hadronic showers may not be fully contained, leading to energy leakage. We will study this effect using the DWC data (wider showers indicate larger leakage) and by comparing results at different energies.

Cross-talk between S and C: Some scintillation light may enter the lead-glass blocks, and some Cherenkov light may be produced in the plastic scintillator. We will calibrate these cross-talk effects using the pure electron beam, where the shower is well understood.

Statistical precision: With 50,000 events per configuration and ~6 momentum settings, we will have ample statistics. The main systematic uncertainties will come from pedestal subtraction, gain calibration, and shower containment corrections.

6. Scientific Significance

Our experiment provides a direct, educational demonstration of one of the most promising detector technologies for the next generation of particle physics experiments. The dual-readout calorimeter is a cornerstone of the IDEA detector proposed for FCC-ee. By testing its fundamental principle with real beam data at CERN, our results will:

- Provide a clear, visual demonstration of the dual-readout principle using C vs S scatter plots.
- Quantify the hadronic energy resolution improvement achievable with the BL4S equipment.
- Measure the energy dependence of the electromagnetic fraction in hadronic showers.
- Connect high school physics to cutting-edge FCC-ee detector R&D, inspiring future physicists.

To our knowledge, no previous BL4S experiment has attempted a dual-readout measurement. The combination of the lead-glass calorimeter (Cherenkov) and plastic scintillators (scintillation) available at CERN T9 provides a unique opportunity to perform this measurement without requiring any custom equipment.

7. Outreach Plan

We plan to share our BL4S experience through several outreach activities:

- School presentations: We will give talks at our school and neighbouring schools, explaining particle physics and our experiment using simple demonstrations (LED light through different media to illustrate Cherenkov vs scintillation).
- Social media: We will document our journey on Instagram and YouTube, from proposal writing to data taking at CERN, making physics accessible to a young audience.
- Publication: We aim to write a report suitable for a physics education journal, contributing to the growing body of BL4S scientific publications.

8. References

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